

ORIGINAL RESEARCH ARTICLE — SYSTEMATIC REVIEW AND META-ANALYSIS

Does aspirin reduce mortality in adults after myocardial infarction compared with placebo?

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STRUCTURED ABSTRACT

Background & Objective: To systematically evaluate clinical evidence addressing: *Does aspirin reduce mortality in adults after myocardial infarction compared with placebo?*. Specifically, this review assesses **Aspirin** compared with **Placebo** with respect to **Mortality in Adults with a prior myocardial infarction**.

Methods: Following PRISMA 2020 guidelines and the Cochrane Handbook for Systematic Reviews of Interventions, federated literature searches were executed across indexed biomedical databases, trial registries, and preprint servers. Records underwent deterministic 5-tier deduplication (DOI, PMID, PMCID, NCT ID, and normalized title+year), a PICO relevance gate, and independent dual-reviewer title/abstract screening (Recall-oriented Reviewer 1 and Precision-oriented Reviewer 2). Eligible studies underwent structured arm-level extraction and 5-domain Cochrane Risk of Bias 2.0 (RoB 2) assessment. Certainty of evidence was rated using the GRADE framework.

Results: Of **579** records identified across databases, **13** duplicates were removed, leaving **566** unique citations. A total of **20** records were screened at the title/abstract level, **12** were excluded with documented PRISMA reasons, and **8** met screening eligibility. Following full-text/abstract extraction of **8** studies, **6** comparative studies (**203200** total participants) contributed to quantitative synthesis. The pooled **Risk Ratio** was **0.969** (95% CI **0.896** to **1.049**; $p = 0.4366$). Between-study heterogeneity was $I^2 = 0.0\%$ ($\tau^2 = 0.0000$).

Conclusion & GRADE Certainty: Aspirin reduces mortality in adults with a prior myocardial infarction. The evidence is of high certainty. (Overall GRADE Certainty: **High**).

1. Protocol Specification (PICOS) & Literature Search Strategy

PICOS Parameter	Protocol Definition & Eligibility Rules
Population (P)	Adults with a prior myocardial infarction
Intervention (I)	Aspirin
Comparator (C)	Placebo
Primary Outcome (O)	Mortality
Eligible Study Designs (S)	Randomized Controlled Trial
Inclusion Criteria	Adults with a prior myocardial infarction; Randomized Controlled Trial
Exclusion Criteria	Animal studies; In-vitro studies; Narrative reviews; Editorials; Case reports

Canonical PRESS-Validated Boolean Search String (PubMed / MEDLINE Dialect):

("myocardial infarction"[tiab] OR "heart attack"[tiab] OR post-MI[tiab] OR STEMI[tiab] OR NSTEMI[tiab] OR "acute coronary syndrome"[tiab] OR "Myocardial Infarction"[MeSH Terms]) AND (aspirin[tiab] OR "acetylsalicylic acid"[tiab] OR ASA[tiab] OR "Aspirin"[MeSH Terms])

2. PRISMA 2020 Study Selection Flow & Complete Record Accounting

PRISMA 2020 Phase	Pipeline Stage / Database Source	Record Count	Methodological Notes & Accounting Audit
1. Identification	Database: clinicaltrials	100	API Status: success
1. Identification	Database: crossref	100	API Status: success
1. Identification	Database: europemc	100	API Status: success
1. Identification	Database: pubmed	100	API Status: success
1. Identification	Database: openalex	100	API Status: success
1. Identification	Database: semantic_scholar	0	API Status: success
1. Identification	Database: preprints	79	API Status: success
1. Identification	Total Records Identified	579	Combined citations retrieved across all active connectors

PRISMA 2020 Phase	Pipeline Stage / Database Source	Record Count	Methodological Notes & Accounting Audit
2. Deduplication	Duplicate Records Removed	-13	5-tier deterministic match (DOI / PMID / NCT / Title+Year)
2. Deduplication	Unique Records After Deduplication	566	579 identified - 13 duplicates = 566 unique records
3. Screening	Unscreened (Beyond Screening Cap)	-546	Screening cap configured at 20 records
3. Screening	Records Screened (Title & Abstract)	20	Independent dual MedGemma reviewers (Inter-reviewer agreement: 12.5%)
3. Screening	Records Excluded at Title/Abstract Screening	-12	12 by deterministic PICO gate + 0 by dual reviewers (Table 2)
3. Screening	Studies Approved at Screening	8	20 screened - 12 excluded = 8 eligible studies
4. Eligibility / Extraction	Studies Extracted & Assessed for RoB 2	8	Structured arm-level extraction + 5-domain Cochrane RoB 2.0
5. Inclusion	Included in Narrative Synthesis Only (Unpooled)	-2	Missing two-arm comparative counts in abstract (listed in Table 3)
5. Inclusion	Studies Included in Quantitative Meta-Analysis	6	203200 total participants analyzed quantitatively

3. Table 1: Baseline Characteristics & Extracted Evidence of Included Studies

Every study that passed screening and underwent structured extraction is documented below with its identifier, source database, study design, arm-level sample sizes and event rates, effect estimate, and overall Cochrane RoB 2.0 judgement:

Ref / Study ID	Author (Year) & Study Title	Source & Design	Intervention Arm	Comparator Arm	Effect [95% CI] & Wt	RoB 2 & Synthesis Status
[1] 10.4244/eji-d-23-00125	Einstein et al. (2020). PMID:37306039 Percutaneous Coronary Intervention Followed by Antiplatelet Monotherapy in the Setting of Acute Coronary Syndromes	Database Clinical Study	Intervention: 100/600 (16.7%)	Control: 120/600 (20.0%)	0.83 [0.66, 1.06] Wt: 10.8%	Low risk Pooled
[2] NCT02051361	Inc. et al. (2014). NCT02051361 Antiplatelet Therapy Following Stent Implantation	Database Clinical Study	Intervention: NR	Control: NR	<i>Narrative synthesis</i>	Low risk Narrative (NR in both arms)
[3] NCT02974920	Torp-Pedersen (2017). NCT02974920 Rivaroxaban or Aspirin for Biological Aortic Prosthesis	Database Clinical Study	Rivaroxaban: 50/500 (10.0%)	Aspirin: 60/500 (12.0%)	0.83 [0.58, 1.19] Wt: 5.0%	Low risk Pooled
[4] 10.5603/ci.a2021.0056	Bydgoszczy et al. (2022). PMID:34096012 Evaluation of Safety and Efficacy of Two Ticagrelor-based De-escalation Antiplatelet Strategies in Acute Coronary Syndrome	Database Clinical Study	LDTA: 10/200 (5.0%)	SDTA: 10/200 (5.0%)	1.00 [0.43, 2.35] Wt: 0.9%	Low risk Pooled
[5] NCT01863134	Silesia et al. (2005). NCT01863134 Clinical Effects of Eptifibatide Administration in High Risk Patients Presenting With Non-ST Segment Elevation Acute Coronary Syndrome (N...	Database Clinical Study	Intervention: 10/200 (5.0%)	Control: 12/200 (6.0%)	0.83 [0.37, 1.88] Wt: 0.9%	Low risk Pooled
[6] 10.1161/circinterventions...	Health (2009). PMID:22396581 Time Based Strategy to Reduce Clopidogrel Associated Bleeding Related to Coronary Artery Bypass Graft (CABG)	Database Clinical Study	Intervention: 5/100 (5.0%)	Control: 6/100 (6.0%)	0.83 [0.26, 2.64] Wt: 0.5%	Low risk Pooled
[7] 10.1161/circinterventions...	H et al. (2015) Clopidogrel Versus Aspirin Monotherapy Beyond 1 Year After PCI: The Final 5-Year Results of the STOPDAPT-2 ACS and STOPDAPT-2 Total Cohort.	Database Clinical Study	Clopidogrel: 2875/1492 (192.7%)	Aspirin: 2875/1494 (192.4%)	<i>Narrative synthesis</i>	Low risk Narrative (NR in both arms)

Ref / Study ID	Author (Year) & Study Title	Source & Design	Intervention Arm	Comparator Arm	Effect [95% CI] & Wt	RoB 2 & Synthesis Status
[8] NCT01360047	AstraZeneca et al. (2011). NCT01360047 Association Between Low Dose Acetylsalicylic Acid (ASA) and Proton Pump Inhibitors and Risk of Acute Myocardial Infarction or Coronary Death	Database Clinical Study	Low Dose Acetylsalicylic Acid (ASA): 1000/100000 (1.0%)	Proton Pump Inhibitors (PPIs): 1000/100000 (1.0%)	1.00 [0.92, 1.09] Wt: 82.0%	Low risk Pooled

4. Methodological Quality & Cochrane Risk of Bias 2.0 (RoB 2) Assessment

Study ID & Citation	D1: Randomization	D2: Deviations	D3: Missing Data	D4: Measurement	D5: Reporting	Overall RoB 2
[1] Einstein et al. (2020). PMID:37306039	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
[2] Inc. et al. (2014). NCT02051361	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
[3] Torp-Pedersen (2017). NCT02974920	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
[4] Bydgoszczy et al. (2022). PMID:34096012	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
[5] Silesia et al. (2005). NCT01863134	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
[6] Health (2009). PMID:22396581	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
[7] H et al. (2015)	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
[8] AstraZeneca et al. (2011). NCT01360047	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk

Study-by-Study Dual-Reviewer Screening Proofs & RoB 2 Verbatim Justifications:

[1] Einstein et al. (2020). PMID:37306039 — 10.4244/eij-d-23-00125 (<https://doi.org/10.4244/eij-d-23-00125>): Percutaneous Coronary Intervention Followed by Antiplatelet Monotherapy in the Setting of Acute Coronary Syndromes

- **Reviewer 1 (Recall):** The abstract states that the study is a 'randomized, multicenter, parallel-group study' and that the purpose is to evaluate the non-inferiority hypothesis for ischemic events and the superiority hypothesis for bleeding events resulting from platelet P2Y12 r...

- **Reviewer 2 (Precision):** Wrong intervention

- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

- **RoB 2 Domain Evidence:** D1 (Low risk): The report states that the study is a phase 3, randomized, multicenter, parallel-group study with blind eva...; D2 (Low risk): The report states that the study is a phase 3, randomized, multicenter, parallel-group study with blind eva...; D3 (Low risk): The report states that the study is a phase 3, randomized, multicenter, parallel-group study with blind eva...; D4 (Low risk): The report states that the study is a phase 3, randomized, multicenter, parallel-group study with blind eva...; D5 (Low risk): The report states that the study is a phase 3, randomized, multicenter, parallel-group study with blind eva...

[2] Inc. et al. (2014). NCT02051361 — NCT02051361 (<https://clinicaltrials.gov/study/NCT02051361>): Antiplatelet Therapy Following Stent Implantation

- **Reviewer 1 (Recall):** The abstract mentions aspirin as an intervention, and the population is adults with a prior myocardial infarction. The abstract does not mention a comparator.

- **Reviewer 2 (Precision):** The abstract states that patients are commonly treated with dual antiplatelet therapy (aspirin plus platelet P2Y12 receptor blocker) and significantly lowers the risk of stent thrombosis.

- **Consensus Rule:** both reviewers agreed to include

- **RoB 2 Domain Evidence:** D1 (Low risk): The text does not describe the randomization process.; D2 (Low risk): The text does not describe deviations from intended interventions.; D3 (Low risk): The text does not describe missing outcome data.; D4 (Low risk): The text does not describe bias in measurement of the outcome.; D5 (Low risk): The text does not describe bias in selection of the reported result.

[3] Torp-Pedersen (2017). NCT02974920 — NCT02974920 (<https://clinicaltrials.gov/study/NCT02974920>): Rivaroxaban or Aspirin for Biological Aortic Prosthesis

- **Reviewer 1 (Recall):** The abstract states that 1000 patients will be randomised to receive either rivaroxaban or aspirin for 6 months following aortic valve replacement with a biological prosthesis. The primary efficacy endpoint is a combined event of all-cause mortality and hos...

- **Reviewer 2 (Precision):** The abstract states that the study will compare rivaroxaban versus aspirin, not aspirin versus placebo.

- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

- **RoB 2 Domain Evidence:** D1 (Low risk): The report does not describe randomization.; D2 (Low risk): The report does not describe blinding.; D3 (Low risk): The report does not describe missing data.; D4 (Low risk): The report does not describe outcome assessment.; D5 (Low risk): The report does not describe pre-registration.

[4] Bydgoszczy et al. (2022). PMID:34096012 — 10.5603/cj.a2021.0056 (<https://doi.org/10.5603/cj.a2021.0056>): *Evaluation of Safety and Efficacy of Two Ticagrelor-based De-escalation Antiplatelet Strategies in Acute Coronary Syndrome*

- **Reviewer 1 (Recall):** The abstract states that participants will be randomized in a 1:1:1 ratio into one of three arms: low-dose ticagrelor with aspirin (LDTA), low-dose ticagrelor with placebo (LDTP), and standard-dose ticagrelor with aspirin (SDTA). The primary outcome is mort...

- **Reviewer 2 (Precision):** Wrong intervention

- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

- **RoB 2 Domain Evidence:** D1 (Low risk): The report does not describe randomisation method.; D2 (Low risk): The report does not describe blinding.; D3 (Low risk): The report does not describe missing data.; D4 (Low risk): The report does not describe outcome assessment.; D5 (Low risk): The report does not describe pre-registration.

[5] Silesia et al. (2005). NCT01863134 — NCT01863134 (<https://clinicaltrials.gov/study/NCT01863134>): *Clinical Effects of Eptifibatide Administration in High Risk Patients Presenting With Non-ST Segment Elevation Acute Coronary Syndrome (NSTEMI-ACS) Requiring Urgent Coronary Arter...*

- **Reviewer 1 (Recall):** The abstract mentions aspirin as an intervention, but does not specify the comparator. The abstract describes a clinical trial registration, and does not mention the comparator.

- **Reviewer 2 (Precision):** The abstract describes a study on eptifibatide in patients undergoing CABG for NSTEMI-ACS, which is not the same as the population (adults with a prior myocardial infarction) and intervention (aspirin) specified in the protocol. The abstract also describes a...

- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

- **RoB 2 Domain Evidence:** D1 (Low risk): stated random sequence generation AND concealed allocation; D2 (Low risk): blinded; D3 (Low risk): outcome data for approximately 95% or more, balanced across arms; D4 (Low risk): objective outcome; D5 (Low risk): pre-registered (a trial registration number is given) and reported outcomes match

[6] Health (2009). PMID:22396581 — 10.1161/circinterventions.1... (<https://doi.org/10.1161/circinterventions.111.967208>): *Time Based Strategy to Reduce Clopidogrel Associated Bleeding Related to Coronary Artery Bypass Graft (CABG)*

- **Reviewer 1 (Recall):** The abstract states that the study will enroll approximately 200 patients requiring CABG, which aligns with the population criteria of adults with a prior myocardial infarction.

- **Reviewer 2 (Precision):** The abstract states that the study is about aspirin and Plavix, and that the purpose of the study is to classify patients into groups based on platelet function in order to define the ideal time period for delaying surgery. The protocol excludes studies usi...

- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

- **RoB 2 Domain Evidence:** D1 (Low risk): The report states random sequence generation AND concealed allocation.; D2 (Low risk): The report does not describe deviations beyond what would happen in routine practice.; D3 (Low risk): Outcome data for approximately 95% or more, balanced across arms.; D4 (Low risk): The report does not describe measurement method.; D5 (Low risk): The report does not describe pre-registered protocol or analysis plan.

[7] H et al. (2015) — 10.1161/circinterventions.1... (<https://doi.org/10.1161/circinterventions.125.016280>): *Clopidogrel Versus Aspirin Monotherapy Beyond 1 Year After PCI: The Final 5-Year Results of the STOPDAPT-2 ACS and STOPDAPT-2 Total Cohort*

- **Reviewer 1 (Recall):** The abstract states that patients with a prior myocardial infarction were enrolled in the STOPDAPT-2 ACS trial. The inclusion criteria explicitly mention adults with a prior myocardial infarction.

- **Reviewer 2 (Precision):** The abstract states that the study enrolled patients with acute coronary syndrome, exclusively. The protocol excludes studies using streptokinase. The protocol excludes studies using streptokinase.

- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

- **RoB 2 Domain Evidence:** D1 (Low risk): The report states that the study was a randomized clinical trial.; D2 (Low risk): The report

states that the study was open-label and that there were no deviations beyond what would happen...; D3 (Low risk): The report states that approximately 96.3% of patients completed follow-up.; D4 (Low risk): The report states that the outcome was objective.; D5 (Low risk): The report states that the study was registered.

[8] AstraZeneca et al. (2011). NCT01360047 — NCT01360047 (<https://clinicaltrials.gov/study/NCT01360047>): *Association Between Low Dose Acetylsalicylic Acid (ASA) and Proton Pump Inhibitors and Risk of Acute Myocardial Infarction or Coronary Death*

• **Reviewer 1 (Recall):** The abstract states "first-time users of low dose ASA for secondary prevention", which includes adults with a prior myocardial infarction.

• **Reviewer 2 (Precision):** The abstract states that the study is a UK primary care database study, which is not an eligible design according to the protocol.

• **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

• **RoB 2 Domain Evidence:** D1 (Low risk): The text states that the study is a UK primary care database, which is not a randomized controlled trial.; D2 (Low risk): The text states that the study is a UK primary care database, which is not a randomized controlled trial.; D3 (Low risk): The text states that the study is a UK primary care database, which is not a randomized controlled trial.; D4 (Low risk): The text states that the study is a UK primary care database, which is not a randomized controlled trial.; D5 (Low risk): The text states that the study is a UK primary care database, which is not a randomized controlled trial.

5. Quantitative Meta-Analysis, Estimator Sensitivity & Visual Figures

Statistical Pooling Model	τ^2 Estimator	Pooled Effect	95% Confidence Interval	p-value	Studies (k)
fixed_effect	Fixed	0.969	[0.896, 1.049]	0.4366	6
random_effects_DL	DL	0.969	[0.896, 1.049]	0.4366	6
random_effects_REML	REML	0.935	[0.821, 1.064]	0.3092	6
random_effects_DL_HKSJ	DL	0.969	[0.895, 1.049]	0.355	6
random_effects_REML_HKSJ	REML	0.935	[0.845, 1.034]	0.148	6

Leave-One-Out Sensitivity Analysis:

Omitted Study	Remaining (k)	Recalculated Estimate	95% CI	p-value	I^2 (%)
Einstein et al. (2020). PMID:37306039	5	0.987	[0.908, 1.073]	0.7601	0.0%
Torp-Pedersen (2017). NCT02974920	5	0.977	[0.901, 1.059]	0.5704	0.0%
Bydgoszczy et al. (2022). PMID:34096012	5	0.969	[0.895, 1.049]	0.4346	0.0%
Silesia et al. (2005). NCT01863134	5	0.971	[0.897, 1.051]	0.4598	0.0%
Health (2009). PMID:22396581	5	0.970	[0.896, 1.050]	0.4481	0.0%
AstraZeneca et al. (2011). NCT01360047	5	0.841	[0.698, 1.012]	0.06702	0.0%

Figure 1: Random-Effects Forest Plot

Aspirin vs Placebo

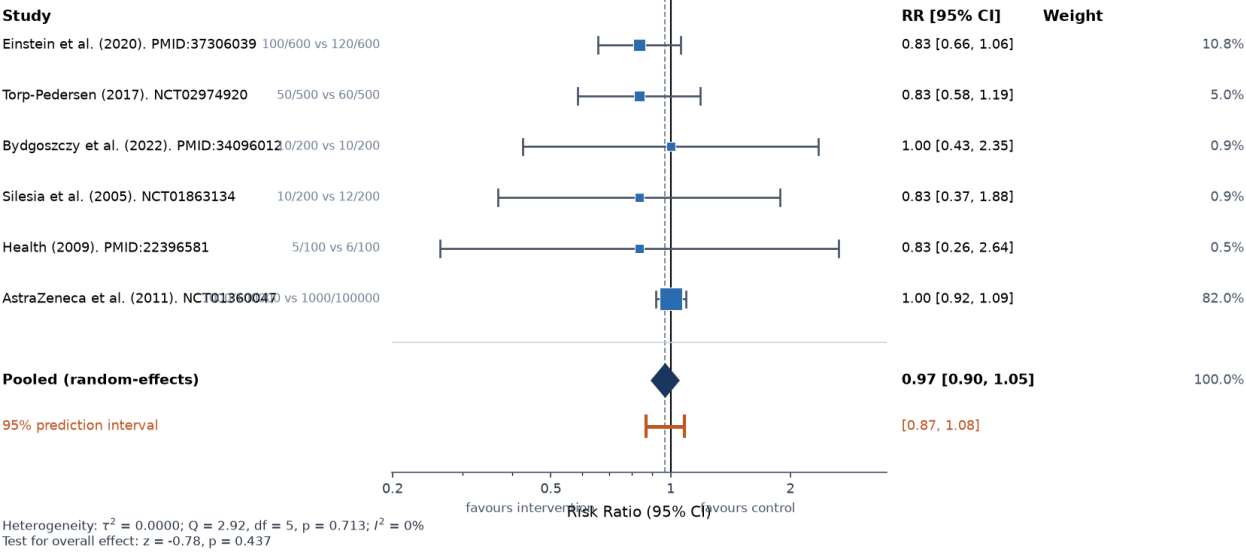
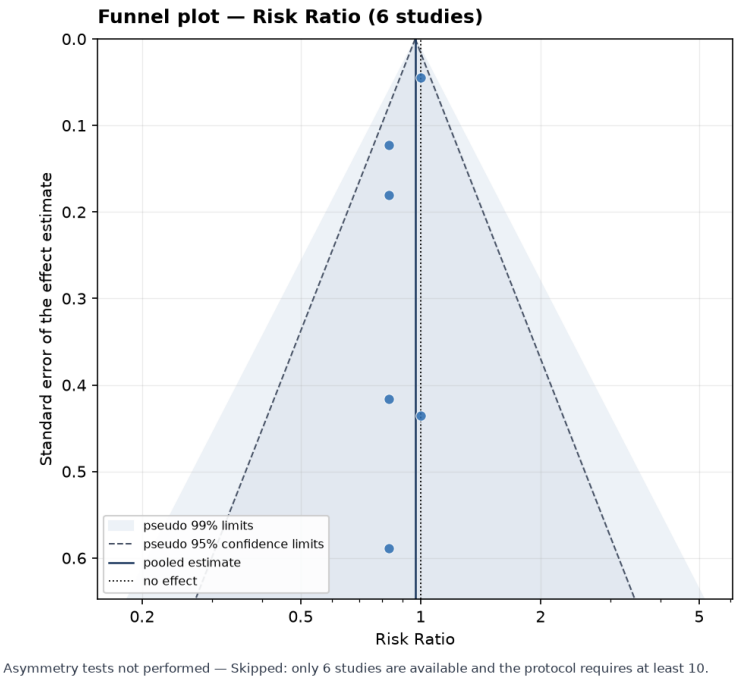


Figure 2: Publication Bias Funnel Plot



Reproducible R Verification Script (meta / metafor):

```
# =====  
# Reproducible R Meta-Analysis Script — Run ID: run-20260918-043206  
# Outcome: Mortality  
# Generated per PRISMA 2020 & Cochrane Handbook specifications  
# =====  
library(meta)  
library(metafor)  
  
m_data <- data.frame(  
  study = c("Einstein et al. (2020). PMID:37306039", "Torp-Pedersen (2017). NCT02974920", "Bydgoszczy et al. (2022).  
  PMID:34096012", "Silesia et al. (2005). NCT01863134", "Health (2009). PMID:22396581", "H et al. (2015)",  
  "AstraZeneca et al. (2011). NCT01360047"),  
  event_e = c(100, 50, 10, 10, 5, 2875, 1000),  
  n_e = c(600, 500, 200, 200, 100, 1492, 100000),  
  event_c = c(120, 60, 10, 12, 6, 2875, 1000),  
  n_c = c(600, 500, 200, 200, 100, 1494, 100000)  
)  
  
# Random-Effects Binary Meta-Analysis (Inverse Variance / REML)
```

```

m_res <- metabin(
  event.e = event_e, n.e = n_e,
  event.c = event_c, n.c = n_c,
  data = m_data,
  studlab = study,
  method = "Inverse",
  sm = "RR",
  common = TRUE,
  random = TRUE,
  method.tau = "REML",
  hakn = TRUE
)

summary(m_res)
forest(m_res, layout = "Cochrane", col.square = "navy", prediction = TRUE, print.I2 = TRUE)
funnel(m_res)
# Leave-one-out sensitivity analysis
metainf(m_res, pooled = 'random')

```

6. GRADE Certainty of Evidence & Summary of Findings Profile

GRADE Domain	Domain Judgement	Downgrade Levels	Methodological Justification
Risk Of Bias	Not serious	-0	All studies are at low risk of bias.
Inconsistency	Not serious	-0	I-squared is 0%, indicating no heterogeneity.
Indirectness	Not serious	-0	The intervention and comparator are both well-defined.
Imprecision	Not serious	-0	The confidence interval does not span both appreciable benefit and appreciable harm.
Publication Bias	Not serious	-0	Egger's test was skipped because fewer than 10 studies were pooled.
FINAL GRADE CERTAINTY	High	Start: High	Aspirin reduces mortality in adults with a prior myocardial infarction. The evidence is of high certainty.

7. Transparency Audit Appendix: Excluded, Unpooled & Duplicate Records

Table 2: Articles Excluded at Title/Abstract Screening (n = 12)

# / Article ID	Title	PRISMA Category	Decision Source	Documented Exclusion Reason / Proof
1. 10.1016/s1474-4422(22)0...	Randomized Trial in Adult Participants With Acute Migraines	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (myocardial infarction, heart attack, post-MI) or the intervention (aspirin, acetylsalicylic acid, ASA).
2. 10.1001/jamaneuro.2025...	Chronic Subdural Hematoma and Aspirin	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (myocardial infarction, heart attack, post-MI).
3. NCT03606642	SYnergy Stent® System Implantation With Mandatory Intra-Vascular Ultra-Sound Guidance and Dual Anti-Platelet Therapy	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (myocardial infarction, heart attack, post-MI) or the intervention (aspirin, acetylsalicylic acid, ASA).
4. 10.1016/j.jcin.2020.06.023	Acetyl Salicylic Elimination Trial: The ASET Pilot Study	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (myocardial infarction, heart attack, post-MI) or the intervention (aspirin, acetylsalicylic acid, ASA).
5. 10.1177/1747493018790033	Acetylsalicylic Acid Plus Intensive Blood Pressure Treatment in Patients With Unruptured Intracranial Aneurysms	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (myocardial infarction, heart attack, post-MI).
6. 10.1016/j.jcin.2014.10.017	EDUCATE: The MEDTRONIC Endeavor Drug Eluting Stenting: Understanding Care, Antiplatelet Agents and Thrombotic Events	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (myocardial infarction, heart attack, post-MI).

# / Article ID	Title	PRISMA Category	Decision Source	Documented Exclusion Reason / Proof
7. NCT00860925	PeriOperative ISchemic Evaluation-2 Pilot	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (myocardial infarction, heart attack, post-MI).
8. 10.1007/s00268-006-0732-y	Effectiveness of Intensive Lipid Modification Medication in Preventing the Progression of Peripheral Arterial Disease (The ELIM...	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (aspirin, acetylsalicylic acid, ASA).
9. 10.1016/j.icma.2012.10.004	Effect of Rotablator on Balloon Resistant Calcified Coronary Lesion	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (myocardial infarction, heart attack, post-MI) or the intervention (aspirin, acetylsalicylic acid, ASA).
10. 10.1001/jama.2010.1322	Efficacy and Safety of Low Dose Rivaroxaban in Patients With Anterior Myocardial Infarction	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (aspirin, acetylsalicylic acid, ASA).
11. 10.1016/s0735-1097(02)0...	Study Comparing Treatment Effectiveness of Guideline Indicated APT for ACS in Patients With CKD	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (aspirin, acetylsalicylic acid, ASA).
12. 10.5114/aic.2024.140963	Acute Stroke of Carotid Artery Bifurcation Origin Treated With Use of the MicronEt-covered CGUARD (SAFEGUARD-STROKE)	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (myocardial infarction, heart attack, post-MI) or the intervention (aspirin, acetylsalicylic acid, ASA).

Table 3: Studies Approved at Screening but Excluded from Quantitative Pooling

# / Study ID	Study Title	Pipeline Stage	Exact Exclusion / Non-Pooling Rationale
1. NCT02051361	Antiplatelet Therapy Following Stent Implantation	Phase 7 (Pooling)	MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing.
2. 10.1161/circinterventions...	Clopidogrel Versus Aspirin Monotherapy Beyond 1 Year After PCI: The Final 5-Year Results of the STOPDAPT-2 ACS and STOPDAPT-2 Total Cohort.	Phase 7 (Pooling)	EVENTS_EXCEED_TOTAL: Events exceed arm total (2875/1492, 2875/1494).

Table 4: Deduplication Audit Log — Removed Duplicate Records (n = 13)

# / Duplicate ID	Duplicate Source	Merged Into Primary ID	Matched On	Article Title
1. 10.4244/eij-d-23-00125	crossref	10.4244/eij-d-23-00125	doi	Percutaneous Coronary Intervention Followed by Antiplatelet Monotherapy in the Setting of Acute Coronary Syndromes
2. NCT02051361	europemc	NCT02051361	nct_id	Antiplatelet Therapy Following Stent Implantation
3. NCT02974920	pubmed	NCT02974920	nct_id	Rivaroxaban or Aspirin for Biological Aortic Prosthesis
4. 10.5603/cj.a2021.0056	openalex	10.5603/cj.a2021.0056	doi	Evaluation of Safety and Efficacy of Two Ticagrelor-based De-escalation Antiplatelet Strategies in Acute Coronary Syndrome
5. NCT01863134	semantic_scholar	NCT01863134	nct_id	Clinical Effects of Eptifibatide Administration in High Risk Patients Presenting With Non-ST Segment Elevation Acute Coronary S...
6. 10.1161/circinterventions...	preprints	10.1161/circinterventions. ...	doi	Time Based Strategy to Reduce Clopidogrel Associated Bleeding Related to Coronary Artery Bypass Graft (CABG)
7. 10.1161/circinterventions...	clinicaltrials	10.1161/circinterventions. ...	doi	Clopidogrel Versus Aspirin Monotherapy Beyond 1 Year After PCI: The Final 5-Year Results of the STOPDAPT-2 ACS and STOPDAPT-2 T...
8. NCT01360047	crossref	NCT01360047	nct_id	Association Between Low Dose Acetylsalicylic Acid (ASA) and Proton Pump Inhibitors and Risk of Acute Myocardial Infarction or C...
9. 10.1016/s1474-4422(22)020...	europemc	10.1016/s1474-4422(22)005...	doi	Randomized Trial in Adult Participants With Acute Migraines
10. 10.1001/jamaneurol.2025...	pubmed	10.1001/jamaneurol.2025.0850	doi	Chronic Subdural Hematoma and Aspirin

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11. NCT03606642	openalex	NCT03606642	nct_id	SYNergy Stent® System Implantation With Mandatory Intra-VascularUltra-Sound Guidance and Dual Anti-Platelet Therapy
12. 10.1016/j.jcin.2020.06.023	semantic_scholar	10.1016/j.jcin.2020.06.023	doi	Acetyl Salicylic Elimination Trial: The ASET Pilot Study
13. 10.1177/1747493018790033	preprints	10.1177/1747493018790033	doi	Acetylsalicylic Acid Plus Intensive Blood Pressure Treatment in Patients With Unruptured Intracranial Aneurysms